

## Does Portfolio Optimization Add Value?

*An Out-of-Sample Analysis of Estimation Risk, Market Regimes, and Transaction Costs*

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### **Research design and empirical evaluation framework**

## **Abstract**

This study investigates whether portfolio optimization adds economic value relative to simple allocation rules once estimation risk, market regimes, portfolio constraints, turnover, and transaction costs are incorporated. The study compares Equal Weight, Global Minimum Variance, Mean-Variance, and Equal Risk Contribution portfolios in a diversified seven-asset ETF universe. Portfolio weights are estimated only from information available before each evaluation month, portfolios are long-only and fully invested, individual positions are capped at 30%, and transaction costs are deducted from realized returns.

The research design evaluates alternative estimation windows, transaction-cost assumptions, covariance shrinkage, risk-aversion parameters, and rebalancing frequencies. Performance is assessed through annualized return and volatility, Sharpe and Sortino ratios, maximum drawdown, tail-loss measures, turnover, concentration, effective holdings, and transaction costs. Moving-block bootstrap procedures are used for uncertainty analysis, including pairwise differences in Sharpe ratios. The design is intended to distinguish genuine out-of-sample portfolio value from effects that arise from estimation noise or implementation assumptions.

The contribution is methodological: it provides a reproducible framework for determining when optimization is economically useful and when apparent improvements are sensitive to modelling or implementation choices.

## **1. Introduction**

The distinction between optimization and investment value is especially important in a finite sample. An optimizer is evaluated against a historical estimate of the joint distribution of returns, whereas an investor acts under uncertainty about that distribution. The relevant empirical object is therefore a decision rule: a mapping from information available at time  $t$  into a portfolio held during the subsequent period. This perspective makes the benchmark choice, information set, constraints, and trading assumptions part of the research design rather than peripheral implementation details.

A further issue is model complexity. Each additional estimated parameter creates another channel through which sampling error can affect portfolio weights. Equal Weight deliberately avoids this problem. Minimum Variance introduces covariance estimation, while Mean-Variance introduces both covariance and expected-return estimation. Equal Risk Contribution also depends on covariance, but imposes a different structural objective. The comparison consequently allows the empirical analysis to ask not only whether optimization works, but whether the incremental information required by more complex rules is reflected in realized outcomes.

The study is also designed to avoid interpreting a backtest as a forecast. Historical evidence can identify patterns within the selected sample, but it cannot establish that a strategy will produce the same outcome in future markets. The strongest claims available from the design concern robustness within the specified historical experiment: whether a result survives alternative windows, costs, covariance estimators, and rebalancing schedules. This is a narrower but more defensible contribution than a claim of universal portfolio superiority.

Portfolio optimization is a central component of modern portfolio theory. Since Markowitz (1952), investors have used estimated expected returns and covariance matrices to translate beliefs about return and risk into portfolio weights. The practical question, however, is not whether an optimizer can produce mathematically efficient weights from a historical dataset. The economically relevant question is whether those weights improve realized investment outcomes when the inputs are estimated, markets change, portfolios are constrained, and trading creates costs.

Optimization is particularly exposed to estimation risk. Expected returns are difficult to estimate precisely, while sample covariance matrices can contain substantial noise. An optimizer can amplify small estimation errors by assigning large differences in portfolio weights to relatively small differences in estimated inputs. A portfolio that is optimal in sample may therefore fail to retain its apparent advantage out of sample.

The literature provides several reasons to treat simple benchmarks seriously. DeMiguel, Garlappi, and Uppal (2009) document the difficulty of improving consistently on equal-weight allocation across many datasets and models. Jagannathan

and Ma (2003) show that portfolio constraints can improve realized performance and may act similarly to covariance shrinkage. Ledoit and Wolf (2003, 2004) demonstrate the importance of improved covariance estimation. These findings motivate an evaluation that incorporates both statistical and implementation uncertainty.

This paper asks whether portfolio optimization adds value after these frictions are taken into account. Four strategies are compared: Equal Weight, Global Minimum Variance, Mean-Variance, and Equal Risk Contribution. Equal Weight provides a transparent benchmark. Minimum Variance uses covariance information without expected-return estimation. Mean-Variance uses both estimated expected returns and covariance. Equal Risk Contribution distributes portfolio risk more evenly across assets.

The study makes five contributions. First, it enforces strict information timing. Second, it compares multiple portfolio construction methods under common constraints. Third, it explicitly deducts transaction costs and measures turnover. Fourth, it evaluates robustness across key modelling assumptions. Fifth, it treats uncertainty and pairwise differences as central parts of the evidence rather than relying on isolated point estimates.

## 2. Research Questions and Hypotheses

The hypotheses are linked to observable quantities. H1 can be examined through pairwise differences in Sharpe ratios, returns, drawdowns, and other performance measures. H2 can be examined by comparing both performance and portfolio characteristics across the 36-, 60-, and 120-month windows. H3 can be examined by tracing net performance across the transaction-cost grid and relating changes in net returns to turnover. H4 can be examined through the difference between sample-covariance and shrinkage specifications, including changes in weight dispersion and realized risk.

H5 requires conditional rather than unconditional evidence. The regime analysis therefore reports strategy statistics separately within Bull, Neutral, and Bear states. The objective is not to choose a regime in advance or to optimize thresholds after seeing returns. The predefined signal creates a transparent partition of the sample. H6 concerns uncertainty and is assessed through bootstrap distributions and pairwise differences rather than through a collection of isolated significance tests.

An important feature of the hypothesis design is that it permits null or mixed findings. If optimization produces no durable improvement, that result is consistent with the research question and can still identify the role of estimation risk or costs. If a strategy improves only under zero costs, the evidence supports a cost-sensitive interpretation. If shrinkage improves stability without materially changing returns, the result informs the statistical mechanism even without producing a large performance gain.

The primary research question is: Does portfolio optimization add value out of sample after estimation risk, market regimes, portfolio constraints, turnover, and transaction costs are taken into account?

The question is decomposed into six subquestions: whether optimized portfolios differ materially from Equal Weight; whether differences survive alternative estimation windows; whether transaction costs alter conclusions; whether covariance shrinkage changes optimization stability; whether relative performance varies across market regimes; and whether observed differences are large relative to statistical uncertainty.

The study proposes five directional hypotheses. H1: optimization may improve risk-adjusted performance relative to Equal Weight under some specifications. H2: the magnitude of any improvement will be sensitive to estimation-window length and parameter uncertainty. H3: transaction costs will reduce the realized advantage of strategies with higher turnover. H4: covariance shrinkage may reduce instability in covariance-dependent optimization. H5: relative strategy performance may vary across market regimes.

These hypotheses are designed so that both positive and negative evidence is informative. The study does not assume that optimization must outperform the benchmark.

## 3. Theoretical Framework

The optimization problem can be understood as a mapping from estimates into decisions. Let  $g(\mu\text{hat}, \Sigma\text{hat})$  denote the portfolio rule. Even if  $\mu\text{hat}$  and  $\Sigma\text{hat}$  are unbiased estimators under a particular statistical model, the nonlinear mapping  $g$  need not preserve unbiasedness in the resulting weights or portfolio performance. This is one reason why the performance of an optimizer cannot be inferred simply from the quality of its input estimators considered separately.

The curvature of the objective also matters. When several assets have similar estimated expected returns and correlations, small changes in estimates can move the optimizer along a relatively flat region of the objective. The resulting weights may change substantially while the estimated objective changes very little. Out-of-sample performance can consequently be sensitive to estimation noise even when the in-sample optimization appears numerically stable.

Constraints alter this mapping. The feasible set is compact under the long-only and maximum-weight restrictions, preventing extreme positions. In practical terms, the constraints act as a regularization mechanism as well as an implementation rule. This means that the empirical comparison is between constrained decision rules, and any interpretation of the results must recognize that the constraints themselves can contribute to stability.

For portfolio weights  $w$ , expected returns  $\mu$ , covariance matrix  $\Sigma$ , and risk-aversion parameter  $\gamma$ , the mean-variance objective can be written as  $\mu'w - (\gamma/2)w'\Sigma w$  subject to portfolio constraints. Global Minimum Variance instead minimizes  $w'\Sigma w$ . Equal Risk Contribution seeks weights whose individual contributions to portfolio risk are approximately equal.

The central statistical problem is that  $\mu$  and  $\Sigma$  are estimated rather than known. Estimation error can be amplified by optimization because the selected weights depend directly on the estimated inputs. The resulting portfolio can therefore be sensitive to the particular historical window used for estimation.

The portfolio decision is treated as a rolling sequence. At time  $t$ , information through  $t-1$  is used to estimate inputs and form weights. Those weights are then applied to the return observed in month  $t$ . This chronology prevents the evaluation-period return from entering the portfolio decision that generated it.

Long-only, full-investment, and maximum-weight constraints make the strategies more implementable and reduce extreme concentration. They also provide a common constraint set that allows the strategies to be compared on a consistent basis.

## 4. Literature Review

The literature also suggests that there is no single source of portfolio-model failure. Expected returns can be difficult to estimate, covariance matrices can be noisy, and the optimization objective can magnify small input differences. Constraints can mitigate some of these effects, while shrinkage can address covariance instability. Transaction costs introduce a separate implementation channel. The present study therefore treats portfolio optimization as an integrated statistical and economic problem rather than as a purely mathematical exercise.

A related theme is benchmark dependence. A strategy can look attractive relative to an inefficient benchmark but less attractive relative to a simple diversified rule. Equal Weight is useful precisely because it establishes a low-parameter benchmark. The relevant question is not whether an optimizer produces an allocation different from Equal Weight; it almost certainly will. The question is whether the additional information encoded in those differences improves realized outcomes enough to justify the associated estimation and trading burden.

The literature on data snooping reinforces the importance of a pre-specified design. Once many strategies, windows, costs, and metrics are examined, isolated favorable results are less persuasive. The robustness matrix in this project is therefore part of the empirical identification strategy. It makes the specification space visible and requires the interpretation to consider the full set of predefined perturbations.

Markowitz (1952) established mean-variance portfolio selection and demonstrated the importance of covariance and diversification. Michaud (1989) later emphasized the sensitivity of optimization to estimation error, framing the issue as a central practical problem rather than a minor statistical detail.

DeMiguel, Garlappi, and Uppal (2009) provide an influential comparison of optimized portfolios and the 1/N strategy. Their findings motivate the use of Equal Weight as a serious benchmark. DeMiguel, Garlappi, Nogales, and Uppal (2009) further investigate generalized portfolio strategies and constraints.

Jagannathan and Ma (2003) show that constraints can reduce risk and improve realized performance. This is directly relevant because the present design imposes long-only, full-investment, and maximum-weight restrictions.

Ledoit and Wolf (2003, 2004) develop covariance estimators designed to improve conditioning and reduce estimation noise. The present robustness analysis uses Ledoit-Wolf shrinkage to determine whether covariance estimation materially affects the conclusions.

Maillard, Roncalli, and Teïletche (2010) formalize Equal Risk Contribution portfolios. Their framework motivates the inclusion of a risk-allocation strategy that differs from both equal capital allocation and mean-variance optimization.

Newey and West (1987) provide heteroskedasticity- and autocorrelation-consistent inference. White (2000) and Harvey, Liu, and Zhu (2016) highlight the danger of data snooping and multiple testing. Politis and Romano (1994) provide foundations for block-bootstrap procedures that preserve dependence structures.

## 5. Data and Investment Universe

The asset universe is intended to create meaningful diversification without introducing an unmanageable number of securities. The seven exposures differ in economic sensitivity: equity assets are linked to corporate earnings and growth,

long-duration Treasury exposure is sensitive to interest rates, gold can respond to real-rate and risk conditions, commodities can reflect inflation and global demand, and real estate combines property exposure with financing and equity-market sensitivity. These differences make the covariance structure economically relevant to the optimization problem.

Using ETFs also makes the portfolio weights interpretable in implementation terms. A 20% weight in SPY, for example, corresponds to an observable traded vehicle rather than an abstract market index. However, the use of ETFs introduces product-level characteristics. Expense ratios, tracking error, fund rebalancing, liquidity, and the timing of distributions can all affect realized returns. The empirical conclusions should consequently remain tied to the chosen securities and sample period.

Data preprocessing is intentionally limited. Month-end sampling reduces the intramonth noise and keeps the portfolio decision frequency consistent with monthly rebalancing. It also means that the study does not model intramonth execution or daily tactical adjustments. The resulting design is best viewed as a medium-frequency strategic allocation experiment.

The empirical universe contains seven ETF proxies: SPY for U.S. equities, EFA for developed-market equities outside the United States, EEM for emerging-market equities, TLT for long-duration U.S. Treasury exposure, GLD for gold, DBC for broad commodities, and VNQ for U.S. real estate investment trusts.

Adjusted market-price data are retrieved through Yahoo Finance using the yfinance Python interface. Prices are converted to month-end observations and returns are computed from consecutive month-end adjusted prices. The analysis therefore uses monthly observations.

The common return panel is aligned across the selected assets. The sample is determined by data availability rather than by choosing dates after observing performance. ETF returns are implementation proxies and can differ from pure asset-class index returns because of fees, tracking differences, fund structure, and market microstructure.

No future information is used to form portfolio weights. For evaluation month  $t$ , the estimation sample ends before  $t$ . The resulting weights are applied to month  $t$ 's return.

## 6. Portfolio Construction Methodology

The four strategies can also be viewed as a sequence of increasing informational assumptions. Equal Weight uses no estimated distributional parameters. Minimum Variance requires covariance estimates. Mean-Variance requires both covariance and expected-return estimates. Equal Risk Contribution requires covariance estimates and a risk-budgeting objective. This ordering is useful because any incremental performance benefit can be considered relative to the statistical information required to generate it.

The use of arithmetic sample means for expected returns is deliberately conservative from a modelling perspective. Sophisticated return forecasts could introduce additional predictive signals, but they would also create more opportunities for model selection and look-ahead bias. The current design isolates the conventional Mean-Variance mechanism and makes its estimation risk transparent.

Equal Risk Contribution illustrates another important distinction: capital diversification and risk diversification are not equivalent. An equal-capital portfolio can allocate much more risk to volatile assets than to low-volatility assets. Conversely, a risk-balanced portfolio may allocate more capital to assets with lower volatility. The realized contribution analysis therefore provides a way to evaluate whether the intended risk-budgeting structure remains visible after the estimation period.

Equal Weight assigns  $1/N$  to each of the seven assets at each rebalance. It requires no estimated expected returns or covariance matrix.

Global Minimum Variance minimizes estimated portfolio variance subject to long-only, full-investment, and 30% maximum-weight constraints.

Mean-Variance maximizes estimated expected return minus a risk penalty. Expected returns are arithmetic means over the rolling estimation window. The baseline risk-aversion parameter is  $\gamma = 3$ , with  $\gamma = 1, 3, 5$ , and  $10$  examined in sensitivity analysis.

Equal Risk Contribution minimizes the dispersion of asset-level risk contributions around their mean. With portfolio volatility  $\sigma_p = \sqrt{w'\Sigma w}$ , the risk contribution of asset  $i$  is proportional to  $w_i(\Sigma w)_i/\sigma_p$ . The optimization seeks approximately equal contributions subject to the common constraints.

Optimization is solved numerically using constrained nonlinear optimization. The 30% maximum position prevents extreme allocations and improves comparability.

## 7. Out-of-Sample Evaluation Design

The rolling design creates a repeated forecasting-like experiment without claiming to forecast returns. At each decision date, the model receives a fixed historical information set and produces weights. The subsequent month is then treated as unseen data. Repeating this process prevents the analysis from relying on one arbitrary split between training and test periods and provides a sequence of investment decisions across changing market environments.

The first observation in the evaluation sample is determined mechanically by the selected estimation window. This matters for comparisons across windows because the usable evaluation periods differ. The robustness analysis should therefore distinguish changes caused by the model from changes caused by the sample available under each window. Reporting the number and dates of observations makes this difference transparent.

Portfolio drift between rebalances is also economically relevant. Under monthly rebalancing, the portfolio is reset every month. Under quarterly rebalancing, weights can move because asset prices change between decisions. The quarterly experiment therefore captures a different balance between responsiveness and trading intensity, rather than simply applying the same weights for three months.

The baseline estimation window contains 60 monthly observations. At each evaluation month, the preceding 60 months are used to estimate portfolio inputs. The resulting weights are applied to the following month's return.

The baseline rebalancing frequency is monthly. Turnover at a rebalance is the sum of absolute changes in portfolio weights relative to the previous portfolio. Transaction costs are proportional to turnover.

The baseline transaction cost is 15 basis points per unit of turnover. Net return equals gross portfolio return less the estimated transaction cost. The initial portfolio formation is not charged an initial turnover cost because no prior portfolio is observed; this convention is recorded explicitly.

The sequence is therefore: estimate from historical data; form constrained weights; observe the next month's asset returns; calculate gross return; calculate turnover; deduct transaction costs; record net return; roll the window forward.

## 8. Transaction Costs and Implementability

Turnover is calculated from changes in portfolio weights rather than from changes in dollar positions alone. This convention is natural for a fully invested portfolio because it measures the fraction of capital that must be traded to transform one allocation into another. It also allows costs to be compared across strategies with different portfolio sizes because the unit is proportional to portfolio value.

The cost grid can be interpreted as a stress test. At zero cost, the analysis isolates the statistical portfolio rule. As the cost rate increases, the experiment progressively incorporates implementation friction. A strategy whose relative performance changes rapidly across the grid has a different economic profile from one whose performance is relatively stable. The break-even cost is therefore a useful descriptive quantity rather than a universal estimate of actual trading expenses.

The cost model is deliberately linear. Real trading costs can be nonlinear, particularly when orders are large relative to market liquidity. Taxes can also depend on investor jurisdiction and holding period. These omissions are limitations, but they are preferable to introducing unsupported security-specific execution estimates that could create false precision.

Transaction costs matter because optimization can generate unstable weights even when gross performance appears attractive. A small gross improvement may not compensate for additional trading activity.

Cost sensitivity uses 0, 5, 10, 15, and 30 basis points. The range is a robustness device rather than a claim about a universal execution cost.

The analysis reports average monthly turnover, average rebalance turnover, and total transaction costs. These measures permit a cost-performance interpretation rather than relying solely on return statistics.

Monthly and quarterly rebalancing are compared. This helps identify whether potential benefits depend on frequent adaptation to changing estimates.

## 9. Estimation Risk and Covariance Shrinkage

Window length changes the bias-variance trade-off in the information set. A short window contains fewer observations and may produce noisy estimates, but it can place more weight on recent relationships. A long window reduces sampling variability but can average across economically different periods. Comparing these windows therefore provides evidence about whether the portfolio rule depends on a particular historical horizon.

Covariance shrinkage is especially relevant because portfolio optimization depends on the inverse-like behavior of covariance estimates even when the objective is expressed without an explicit matrix inverse. Small changes in correlations and variances can affect the optimizer's preferred trade-offs. Improving the conditioning of the covariance estimate can therefore alter both numerical stability and economic allocations.

The shrinkage experiment should be examined through both weights and returns. If the shrinkage estimator produces smoother weights but similar realized performance, its principal benefit may be stability rather than return enhancement. If it changes realized drawdowns or Sharpe ratios, the evidence suggests that covariance estimation itself materially affects the investment outcome.

Three estimation windows are evaluated: 36, 60, and 120 months. A shorter window can respond more quickly to changing conditions but contains fewer observations. A longer window provides more observations but can adapt more slowly.

Ledoit-Wolf shrinkage is used as a covariance robustness specification. It trades some sample flexibility for improved conditioning and potentially lower estimation error. Because Minimum Variance, Mean-Variance, and Equal Risk Contribution depend on covariance estimates, this experiment directly tests a key source of optimization instability.

If shrinkage changes portfolio weights substantially but leaves conclusions stable, the evidence is more robust to covariance estimation. If conclusions change materially, covariance estimation becomes an important qualification.

## 10. Market-Regime Analysis

The regime rule uses only a broad market signal and deliberately avoids endogenous strategy information. SPY is chosen because it provides a simple and widely interpretable representation of U.S. equity-market conditions. The trailing 12-month horizon smooths short-term noise, while the  $\pm 10\%$  thresholds create three states with a neutral middle region.

The one-month lag is essential. Without the lag, the return of the evaluation month could enter the trailing signal and partially determine the label assigned to that same return. Such leakage would bias the conditional analysis. The lagged construction ensures that regime membership is known before the performance observation is realized.

Conditional statistics should be interpreted with sample-size awareness. A Bear regime may contain fewer observations than a Neutral regime, and its annualized statistics can therefore be sensitive to individual months. The regime analysis is consequently descriptive and should complement rather than replace the full-sample comparison.

Market conditions may affect relative strategy performance. Each evaluation month is classified using information available before that month. The regime signal is based on the trailing 12-month SPY return observed at the prior month-end.

A lagged trailing return above  $+10\%$  defines Bull, below  $-10\%$  defines Bear, and intermediate values define Neutral. The signal is shifted so that the evaluation month's return cannot affect its own classification.

Regime analysis reports observations, annualized return, Sharpe ratio, and maximum drawdown for each strategy within each regime. Small regime samples are interpreted cautiously.

## 11. Performance Measurement

Compounding is used for annualized return because investment wealth accumulates multiplicatively. Arithmetic average returns would not describe the realized growth of a portfolio over the full evaluation period. Volatility, by contrast, is based on dispersion around the sample mean and is annualized through the standard monthly scaling convention.

The Sharpe ratio provides a standardized risk-adjusted measure but has limitations. It assumes that mean and standard deviation summarize the relevant risk-return trade-off, while portfolio returns can exhibit skewness, kurtosis, serial dependence, and drawdowns. The study therefore reports Sharpe alongside downside and tail measures rather than treating it as a sufficient statistic.

Effective holdings are particularly useful when interpreting optimized weights. Two portfolios can have the same maximum weight but different overall concentration. The inverse Herfindahl-style measure captures this distinction and gives a compact representation of diversification. It should be read together with the full weight history because any summary measure necessarily loses temporal information.

Annualized return is calculated from compounded monthly net returns. Annualized volatility is monthly standard deviation multiplied by  $\sqrt{12}$ . Sharpe ratios are calculated using a zero annual risk-free-rate convention and a 2% annual sensitivity.

Maximum drawdown is calculated from the cumulative wealth index. Monthly 5% Value-at-Risk is the empirical fifth percentile. Conditional Value-at-Risk is the mean return among observations at or below that percentile.

Sortino performance is reported using downside deviation. Skewness and excess kurtosis provide supplementary information about distributional shape.

Implementation measures include turnover, total transaction costs, maximum portfolio weight, and effective holdings calculated as the inverse of the sum of squared weights.

## 12. Statistical Inference and Uncertainty

The moving-block bootstrap is chosen because monthly portfolio returns can contain local dependence. Resampling individual months independently would impose an iid structure that may not reflect the data. Six-month blocks are a pragmatic compromise: they preserve medium-term local sequences while still allowing many possible resampled paths.

Bootstrap intervals should not be interpreted as a substitute for economic reasoning. A wide interval communicates uncertainty, while a narrow interval indicates greater precision within the bootstrap design. Neither establishes that historical performance will persist in a different market environment. The intervals are therefore evidence about the reliability of the sample statistic, not guarantees about future returns.

Pairwise differences are the most directly relevant uncertainty objects. If the research question asks whether strategy A differs from strategy B, the statistic of interest is their relative performance. Reporting only separate intervals around each strategy can obscure this point because overlapping or non-overlapping marginal intervals do not provide a complete test of the paired difference.

A difference in point-estimate Sharpe ratios is not automatically evidence of a reliable performance difference. The baseline uncertainty analysis uses a moving-block bootstrap with six-month blocks to preserve local dependence in monthly returns.

Bootstrap intervals are calculated for individual strategy Sharpe ratios and pairwise Sharpe-ratio differences. Pairwise differences directly address the question of whether one strategy's observed performance differs from another's.

The research design treats window length, costs, risk aversion, covariance estimation, and rebalancing as pre-defined robustness dimensions. This reduces the risk that specifications are selected after observing favorable results.

Statistical significance and economic significance are distinct. The interpretation therefore considers effect size, uncertainty, turnover, costs, and robustness jointly.

## 13. Robustness Matrix

The robustness matrix functions as a pre-registered-style specification map. It defines the principal dimensions along which reasonable researchers might disagree: historical window, cost assumption, risk-aversion parameter, covariance estimator, and rebalancing frequency. Because these choices are fixed before the empirical narrative is written, the analysis reduces the temptation to select a specification based on favorable results.

Robustness should be assessed qualitatively as well as numerically. A strategy whose Sharpe ratio changes from 0.60 to 0.58 across two specifications may be broadly stable, while a strategy whose relative position changes because of a small cost adjustment may be economically fragile. The appropriate unit of interpretation is therefore the pattern across the matrix, not the number of decimal places in one cell.

The matrix also provides a natural basis for sensitivity figures. Heatmaps or line charts can show net performance as costs rise, performance as window length changes, and weight concentration under alternative covariance estimators. These visualizations can make specification dependence easier to audit than a long sequence of isolated tables.

The baseline specification is a 60-month window, 15-basis-point cost,  $\gamma = 3$ , monthly rebalancing, sample covariance, long-only constraints, full investment, and a 30% maximum weight.

Window robustness uses 36, 60, and 120 months. Cost robustness uses 0, 5, 10, 15, and 30 basis points. Mean-Variance risk aversion uses  $\gamma = 1, 3, 5$ , and 10. Covariance robustness compares sample covariance with Ledoit-Wolf shrinkage. Rebalancing robustness compares monthly and quarterly schedules. Risk-free-rate sensitivity compares 0% and 2% annual assumptions.

A result that survives materially different assumptions provides stronger evidence than a result confined to one specification. A specification-sensitive result should be described accordingly.

## 14. Research Integrity and Reproducibility

Reproducibility is treated as an empirical requirement rather than a software convenience. A reader should be able to identify the source of the data, the portfolio rules, the information timing, the cost assumptions, and the exact computational procedure used to generate the reported statistics. Separating generated output from manuscript prose makes this chain easier to audit.

The repository also separates validation from empirical execution. Continuous integration verifies that the source code compiles and that required research documents exist. The empirical pipeline itself generates results when executed in an environment with the required data access. This distinction prevents a successful software build from being misrepresented as evidence that the financial results have been reproduced.

A strong integrity protocol also requires documenting negative results and computational failures. If a solver fails for a particular window or if a factor dataset is unavailable for part of the sample, the issue should be recorded. Omitting inconvenient specifications creates a more serious methodological problem than reporting that an analysis could not be completed.

The repository separates methodology, literature, source code, data documentation, results, and the manuscript. The computational pipeline is designed to generate empirical outputs from the underlying data rather than embedding manually typed performance numbers.

Provenance records include the Python environment, platform information, ticker universe, sample period, and baseline parameters. Continuous integration checks required research files and verifies that the principal source file compiles.

Empirical claims should be traceable to generated output files. Presentation quality is not treated as evidence. This is particularly important in an independent research project intended for academic evaluation.

The research protocol is therefore designed so that final numerical findings can be incorporated after execution without changing the methodological rules that determine the test.

## 15. Empirical Results Structure

The baseline table should be the anchor of the empirical section. It should provide enough information for a reader to understand both return and risk before moving to robustness. The cumulative wealth figure should use exactly the same net-return series, avoiding the common problem of presenting a chart based on a slightly different calculation from the table.

Robustness tables should preserve comparability. If the evaluation sample changes because a longer estimation window requires more history, that fact should be shown in the table. Similarly, if a cost assumption changes only the transaction-cost deduction while leaving gross returns unchanged, both gross and net measures should be distinguished where useful.

The interpretation should avoid data-mining language. A favorable result in one cell of a large matrix is not sufficient to establish a robust effect. The reader should be shown whether the result persists across nearby specifications and whether the associated uncertainty interval is consistent with an economically meaningful difference.

The completed empirical section should begin with a baseline performance table containing return, volatility, Sharpe ratio, Sortino ratio, drawdown, tail-loss measures, turnover, transaction costs, and effective holdings. A cumulative net-wealth figure should accompany the table.

Subsequent tables should report estimation-window and transaction-cost sensitivity, followed by covariance shrinkage, risk-aversion, and rebalancing results. Regime results should then show conditional performance across Bull, Neutral, and Bear states.

The final statistical table should present bootstrap intervals for individual Sharpe ratios and pairwise Sharpe differences. Interpretation should proceed from descriptive results to robustness, uncertainty, and economic significance.

## 16. Interpretation Framework

A useful interpretation starts with descriptive facts and then moves toward explanation. First identify the realized performance pattern. Next examine whether the pattern is associated with differences in volatility, drawdown, concentration, or turnover. Then test whether it survives costs and alternative estimation choices. Finally assess statistical uncertainty. This sequence reduces the risk that an attractive headline statistic determines the interpretation before the underlying mechanisms are examined.

The framework also distinguishes robustness from magnitude. A very small effect that appears under every specification is different from a large effect that appears only under one combination of assumptions. The former may be statistically

robust but economically modest; the latter may be economically interesting but specification-sensitive. Both dimensions should be visible in the final discussion.

Interpretation should remain within the historical scope of the experiment. A result can support the statement that a strategy performed differently from another strategy in the selected sample under the stated assumptions. It cannot, by itself, establish that the same relationship will hold in future periods. This distinction is particularly important for investment-related research.

The study does not treat portfolio construction as a contest with a universal winner. It evaluates four dimensions: performance, risk, implementation cost, and robustness.

Performance concerns realized return and risk-adjusted return. Risk concerns volatility, drawdowns, and tail losses. Implementation concerns turnover, concentration, and transaction costs. Robustness concerns stability across alternative assumptions.

If an optimizer improves the baseline Sharpe ratio but loses that improvement at higher costs, the evidence should be described as cost-sensitive. If covariance shrinkage changes weights but not the conclusion, the result is robust to that estimation choice. If performance differs by regime, the conclusion should report the conditional pattern rather than extrapolate universally.

This approach keeps conclusions tied to observed evidence and avoids turning small historical differences into broad claims about future investment performance.

## 17. Limitations

The limited universe also constrains generalization across investor objectives. An institution with liabilities, tax constraints, leverage limits, or liquidity requirements may face a different optimization problem. The present study intentionally abstracts from these features to create a common experiment across strategies.

The monthly frequency introduces a further trade-off. It reduces microstructure noise and makes the portfolio process manageable, but it may miss short-lived risk events and intramonth drawdowns. A strategy that appears stable at month-end could experience substantial interim volatility that is not captured by the return series.

Finally, the historical period may contain unusual episodes that influence aggregate statistics. The robustness framework can reveal whether results are concentrated in particular regimes or subperiods, but it cannot remove the fundamental dependence of a backtest on the realized history. The appropriate response is transparent scope rather than overgeneralization.

The seven-asset ETF universe does not represent the full opportunity set available to investors. Results therefore apply to the specified universe and implementation proxies.

Historical backtests are sample-dependent. Structural breaks, changing market regimes, and changing asset characteristics limit the extent to which historical results can establish future performance.

The proportional transaction-cost model does not capture every execution friction, including market impact, taxes, financing costs, and investor-specific trading conditions.

Mean-Variance results depend on the expected-return estimator, covariance estimator, objective, risk-aversion parameter, constraints, and optimization method. Equal Risk Contribution also depends on covariance estimation and the numerical objective.

Monthly observations provide fewer effectively independent observations than the raw sample length suggests. Bootstrap and HAC procedures can improve inference but cannot create information absent from the sample.

## 18. Conclusion

The study's central contribution is therefore a disciplined comparison rather than a new optimization algorithm. By holding the investment universe, constraints, evaluation frequency, and information-timing rules constant, the analysis focuses attention on what changes when portfolio construction becomes more model-dependent. The design makes it possible to distinguish an optimizer's theoretical objective from the realized consequences of estimating and implementing it.

The most informative empirical outcome may be a pattern of trade-offs rather than a single dominant statistic. For example, an optimization rule may reduce volatility but increase turnover; shrinkage may stabilize weights without

materially changing return; or a gross return difference may disappear after costs. Each pattern answers a different part of the research question and can be interpreted within the same framework.

The final empirical conclusion should consequently be proportional to the evidence generated by the pipeline. The paper is strongest when every important numerical statement can be traced to a reproducible calculation and when the discussion acknowledges the conditions under which the observed pattern changes. This is the standard applied throughout the research design.

This research develops a reproducible framework for evaluating whether portfolio optimization adds value when exposed to estimation uncertainty, changing market conditions, portfolio constraints, turnover, and transaction costs.

The central methodological point is that optimization should not be judged solely by an in-sample objective or a single out-of-sample statistic. Economic value depends on whether the resulting allocation remains effective when estimates are noisy and implementation is costly.

The empirical framework can distinguish persistent improvement from cost-sensitive, regime-dependent, or specification-sensitive improvement. Each outcome is informative because it addresses a different component of the research question.

For investment practice, the framework emphasizes that portfolio construction methods should be assessed from estimation through implementation. For an MSc-level research setting, it demonstrates transparent chronology, reproducible computation, explicit uncertainty, and disciplined interpretation.

## 21. Technical Implementation and Data Validation

A reproducible empirical study requires validation at several layers. The first layer is source-data validation. The downloaded price panel should be checked for duplicate dates, missing asset columns, non-monotonic timestamps, and observations that are inconsistent with the expected monthly frequency. These checks do not determine whether the data are economically correct, but they prevent simple data-engineering errors from becoming financial conclusions.

The second layer is return construction. Adjusted month-end prices are ordered chronologically and monthly returns are calculated from consecutive observations. The transformation should be deterministic: the same input price panel and the same code should produce the same return panel. The pipeline therefore saves the resulting monthly return dataset as an output artifact so that subsequent portfolio calculations can be inspected without repeatedly downloading data.

The third layer is alignment. All seven assets must refer to the same evaluation months. A common panel is preferable to silently changing the universe through time because strategy comparisons should be made on the same underlying observations. The cost of this decision is that the usable sample can become shorter if one asset has missing observations. That trade-off is documented rather than hidden.

The fourth layer is information timing. The rolling estimation window is constructed by positional slicing of observations strictly before the evaluation index. This implementation detail is important because a one-period indexing error can invalidate the entire backtest. The source code should therefore make the timing convention explicit and the provenance output should record the baseline window and evaluation rule.

The fifth layer is output validation. Generated tables should be checked for expected strategy names, observation counts, finite statistics, and plausible weight constraints. Portfolio weights should sum to one within numerical tolerance and no weight should fall outside the prescribed bounds. These checks are simple but important because numerical optimization can fail silently if its outputs are not inspected.

The validation framework also separates data errors from model errors. A missing observation is a data issue; an infeasible optimization problem is a computational issue; and an unexpectedly high turnover series may be a model characteristic rather than an error. Keeping these categories separate improves the auditability of the final research record.

## 22. Covariance Estimation in Detail

The sample covariance matrix is estimated from the rolling monthly return observations. For assets  $i$  and  $j$ , the sample covariance measures the joint variation of their returns around their respective sample means. The resulting matrix contains both volatility information on the diagonal and co-movement information off the diagonal. Portfolio variance is then calculated as  $w'\hat{\Sigma}w$ .

Covariance estimation is central because diversification benefits depend on co-movement. An asset with high individual volatility can still reduce portfolio risk if its correlation with the rest of the portfolio is sufficiently low. Conversely, assets that appear diversified by label can provide limited diversification if their returns become highly correlated during stress periods. The optimizer attempts to exploit these estimated relationships, which is precisely why estimation error matters.

A rolling covariance matrix also creates a time-varying investment rule. Even if the optimization objective is fixed, the weights can change because the estimated correlations and volatilities change. This makes turnover partly endogenous to the statistical estimation process. The turnover diagnostics can therefore be interpreted as a practical manifestation of covariance instability.

Ledoit-Wolf shrinkage modifies the covariance estimate by combining sample information with a structured target. The resulting estimator is generally better conditioned than a raw sample covariance matrix in settings where estimation noise is important. In this seven-asset application, dimensionality is not extreme, but shrinkage remains informative because the objective is sensitive to the estimated covariance structure even in a relatively small universe.

The shrinkage comparison should not be framed as a contest between an old and a new estimator. Its role is diagnostic. If sample covariance and shrinkage produce similar performance and similar weights, covariance estimation may not be the dominant source of instability in the experiment. If they produce materially different allocations or risk outcomes, covariance estimation becomes an important part of the explanation.

A useful extension would examine the condition number of the covariance matrix through time. Very large condition numbers indicate that the matrix contains directions with very different estimated variances and can be numerically sensitive. Recording such diagnostics would provide an additional link between statistical conditioning and portfolio behavior.

## 23. Mean-Variance Estimation and Parameter Sensitivity

Mean-Variance optimization introduces a second estimated object beyond covariance: the expected-return vector. The baseline expected return for each asset is the arithmetic mean of its monthly returns within the rolling estimation window. This estimator is transparent and easy to reproduce, but it is statistically noisy. The research design therefore treats the Mean-Variance strategy as an especially demanding test of whether estimated return information adds value.

The expected-return estimate enters the objective linearly, while variance enters through the quadratic penalty. Consequently, the relative scale of  $\mu$  and  $\Sigma$  is economically important. The risk-aversion parameter  $\gamma$  determines how strongly the optimizer penalizes variance relative to estimated return. Because the units of the inputs are fixed by the monthly return convention, the gamma sensitivity grid is necessary for interpreting the baseline result.

The  $\gamma = 1, 3, 5$ , and  $10$  specifications provide a structured sensitivity analysis. Lower gamma places relatively more emphasis on estimated return, while higher gamma places greater emphasis on estimated risk. The exact economic interpretation should be made cautiously because the historical sample mean is not a forward-looking expectation in the strict asset-pricing sense.

The Mean-Variance strategy can also be understood as a stress test for model complexity. If it performs well only under a narrow gamma value, the result may indicate objective sensitivity. If it remains broadly stable across the grid, the evidence is more consistent with a robust portfolio rule. Neither pattern establishes future performance, but both reveal how dependent the historical result is on the specification.

A further diagnostic is the dispersion of estimated expected returns. If the cross-sectional means are very close, large weight differences generated by the optimizer are unlikely to reflect large differences in estimated return and may instead arise from covariance interactions. If the means are highly dispersed, the optimizer may intentionally concentrate in assets with higher estimated returns, making concentration and turnover especially important to report.

For this reason, Mean-Variance results should never be presented without the associated weight diagnostics. A headline performance statistic cannot reveal whether the strategy's outcome was generated by modest allocation changes across the universe or by repeated near-boundary positions. The latter would carry a different interpretation and implementation burden.

## 24. Equal Risk Contribution Mechanics

Equal Risk Contribution is based on risk allocation rather than expected return maximization. Let  $\sigma_p$  denote portfolio volatility. The contribution of asset  $i$  to total portfolio volatility is proportional to  $w_i(\Sigma w)_i/\sigma_p$ . If all contributions are equal, each asset contributes approximately the same amount of estimated portfolio risk, even though the amount of capital invested in each asset can differ substantially.

This construction creates an important contrast with Equal Weight. Equal Weight treats each asset symmetrically in capital space. Equal Risk Contribution treats each asset symmetrically in estimated risk space. If assets have different volatilities, a lower-volatility asset may receive a larger capital allocation under risk balancing than under Equal Weight.

The covariance matrix remains central because marginal risk depends on the complete covariance structure rather than on individual volatility alone. Consequently, Equal Risk Contribution can also be affected by estimation noise. A change in estimated correlation can alter the risk contribution of an asset and therefore its target weight.

The numerical objective minimizes the dispersion of risk contributions around their mean. This formulation is convenient for a constrained numerical implementation, but it is important to distinguish an approximate numerical solution from an exact analytical risk-parity solution. Convergence diagnostics and the realized dispersion of contributions should therefore be recorded.

Ex-post risk contributions provide a useful test of the distinction between estimated and realized risk. The portfolio is constructed using historical covariance, but its realized returns are generated by future covariance relationships. If risk contributions change materially out of sample, that is direct evidence of the uncertainty involved in treating historical covariance as known.

The strategy also provides a useful bridge between optimization and portfolio management practice. Risk budgeting can be interpreted as a rule for controlling concentration in risk rather than capital. The empirical analysis therefore examines whether such a structure produces stable implementation characteristics and whether its benefits, if any, persist after transaction costs.

## 25. Turnover, Portfolio Drift, and Rebalancing

Turnover is not merely a cost input; it is a diagnostic of portfolio stability. A strategy with rapidly changing weights is responding strongly to new information or to changes in estimated parameters. That response can be economically desirable if the information is reliable, but it can be costly or harmful if the changes are primarily sampling noise.

Under monthly rebalancing, the target portfolio is recalculated at every decision date. The portfolio is then exposed to the next month's realized returns. At the following rebalance, the new target is compared with the previous portfolio. This comparison captures both changes in the optimizer's target weights and the effect of portfolio drift caused by relative asset returns.

Quarterly rebalancing changes this process. Between rebalance dates, the portfolio is allowed to drift, reducing the frequency with which estimated information is translated into trades. The trade-off is that the portfolio may deviate further from its target allocation before the next rebalance. This is economically meaningful because rebalancing frequency affects both costs and risk exposure.

The cost-performance analysis should therefore distinguish gross and net returns. Gross performance answers the statistical question of what the allocation rule generated before implementation costs. Net performance answers the more practical question of what remains after the specified trading friction. Comparing the two reveals the economic price of adaptation.

A useful supplementary analysis is the relationship between turnover and subsequent returns. This should be treated descriptively rather than as a causal test. High turnover may occur because the optimizer identifies a genuine change in opportunity, because estimates become unstable, or because asset prices move sharply. The data alone do not identify the mechanism without a separate structural model.

The rebalance comparison also illustrates why implementation assumptions can affect apparent model performance. A portfolio strategy is not fully defined by its weight formula. The frequency at which the formula is applied is part of the strategy. Two investors using the same optimizer but different rebalance schedules can therefore experience materially different realized portfolios.

## 26. Wealth Index, Drawdowns, and Path Dependence

Portfolio evaluation is inherently path-dependent because wealth compounds through time. Two strategies can have the same arithmetic average monthly return and different terminal wealth if their return paths differ. The cumulative wealth index therefore provides information that annualized averages alone cannot provide.

Drawdown analysis adds another path-dependent perspective. Maximum drawdown records the largest peak-to-trough loss in the cumulative wealth series. It does not measure how frequently losses occur or how long recovery takes, so the final analysis should inspect the timing of drawdowns rather than relying on the maximum alone.

The interaction between drawdown and regime is particularly informative. A strategy may have moderate full-sample volatility but experience concentrated losses during Bear periods. Conversely, a strategy can have relatively higher average

volatility while limiting certain drawdown episodes through diversification. Conditional analysis can help distinguish these profiles.

Tail measures complement drawdowns by focusing on the distribution of monthly outcomes. The empirical 5% VaR identifies a threshold associated with the lower tail, while CVaR summarizes the average outcome within that tail. These statistics are sensitive to sample size, so they should be reported as descriptive measures rather than precise estimates of extreme future risk.

Skewness and kurtosis provide additional context. A portfolio with negative skewness may experience occasional large losses that are not fully captured by average volatility. Excess kurtosis indicates heavier tails relative to a normal distribution. These characteristics matter when comparing strategies that have similar Sharpe ratios but different downside profiles.

The wealth index should begin from a common value for every strategy and use net returns after transaction costs. This ensures that the visual comparison reflects the same economic assumptions as the performance tables. Mixing gross and net series would make the figure misleading even if the underlying return calculations were correct.

## 27. Subperiod and Stability Analysis

Full-sample statistics can conceal changes through time. A strategy may perform strongly during one historical period and weakly during another, while the aggregate statistic averages the two. Subperiod analysis is therefore useful as a descriptive robustness check even when it is not treated as a separate hypothesis test.

The primary regime analysis already conditions performance on Bull, Neutral, and Bear states. A complementary subperiod analysis can examine broad chronological segments, such as early, middle, and late portions of the evaluation sample. The purpose is to determine whether the central pattern is concentrated in one historical era rather than to select favorable dates after seeing the results.

Chronological subperiods should be defined mechanically. For example, the sample can be divided into approximately equal-length segments based on dates or observation counts. The exact boundaries should be recorded before interpretation. This prevents arbitrary subperiod selection from becoming another form of data mining.

Stability can also be examined through rolling performance measures. A rolling 36-month Sharpe ratio or volatility estimate can show whether a strategy's behavior is persistent or episodic. Rolling diagnostics should be interpreted cautiously because overlapping windows create dependence between adjacent estimates.

The combination of regime and subperiod analysis provides two different conditioning perspectives. Regimes ask whether performance differs under a predefined market-state indicator. Chronological subperiods ask whether the result is concentrated in a particular historical era. If both point to the same period, the evidence suggests concentration; if they differ, the mechanism may be more complex.

These analyses do not increase the amount of information in the sample. They divide it. Consequently, conditional estimates can be noisier than full-sample estimates. Their value lies primarily in detecting concentration and instability, not in producing additional opportunities to identify favorable results.

## 28. Multiple Testing and Data-Snooping Risk

The research evaluates many related specifications. This is necessary for robustness, but it creates a statistical issue: the more comparisons are examined, the greater the chance that at least one will look unusually favorable by chance. This problem is often described as data snooping or multiple testing.

The project addresses the issue first through design. The main robustness dimensions are defined before the empirical interpretation: estimation windows, transaction costs, gamma values, covariance estimator, rebalancing frequency, and risk-free sensitivity. This means that the final paper can report the complete specification space rather than presenting only the configurations that support a preferred narrative.

Second, the paper distinguishes exploratory from confirmatory evidence. A pattern discovered in a robustness table can motivate further research, but it should not automatically be presented as a confirmed effect. Stronger claims require persistence across nearby specifications and appropriate uncertainty analysis.

Third, pairwise performance comparisons are treated as the principal relative objects. If four strategies are compared, there are six pairwise combinations. The bootstrap output reports these differences explicitly. This is more informative than identifying the strategy with the largest point estimate and treating the difference as self-evident.

Formal multiple-testing corrections can be incorporated in future versions, including superior predictive ability procedures or model-confidence-set methods. The present framework does not claim that its bootstrap intervals eliminate the multiple-comparison problem. Instead, it makes the problem visible and limits interpretive overreach.

A related concern is researcher degrees of freedom. Decisions about the universe, window, costs, regime thresholds, and performance measures can all affect the result. The project reduces this flexibility by documenting these choices in the methodology before empirical findings are incorporated. The remaining choices should be disclosed rather than presented as inevitable.

## 29. Economic Significance and Decision-Relevant Evidence

Statistical evidence and economic significance answer different questions. Statistical evidence concerns the precision of an estimated difference under a specified sampling framework. Economic significance concerns whether the magnitude of that difference is meaningful after considering costs, risk, concentration, and implementation complexity.

For portfolio optimization, this distinction is especially important because the potential improvement over Equal Weight may be modest. If an optimizer produces a small increase in annualized return while materially increasing turnover, the net benefit may be economically limited. Conversely, a modest return difference combined with a meaningful reduction in drawdown could be relevant for an investor with strong downside sensitivity.

The paper therefore avoids collapsing performance into a single ranking. Instead, the evidence profile includes return, volatility, Sharpe, Sortino, drawdown, tail loss, turnover, transaction costs, and effective holdings. This multidimensional reporting allows a reader to apply an objective that is appropriate to their own investment constraints without the research imposing one universal utility function.

Break-even transaction costs provide a particularly intuitive economic measure. If a gross-performance difference disappears at a relatively small cost assumption, the result is highly implementation-sensitive. If the difference persists across the entire cost grid, the evidence is more robust to the specified trading friction.

Similarly, effective holdings can reveal whether an apparent performance gain comes with concentration. A strategy that repeatedly reaches the maximum allowed position in one or two assets is exposed to a different set of risks from a strategy with broadly distributed weights. Concentration is therefore part of the economic interpretation, not merely a portfolio-statistics footnote.

Ultimately, the empirical evidence should support conditional statements. The research can establish what occurred within the historical experiment and under which assumptions. It cannot convert those observations into a guarantee of future returns or into a universal prescription for investors with different constraints.

## 30. Extension to Formal Model Comparison

The current framework compares strategies primarily through descriptive performance, pairwise Sharpe differences, robustness analysis, and bootstrap uncertainty. A natural extension would formalize the comparison using a superior predictive ability framework. Such a procedure is designed for settings in which multiple candidate models are evaluated and one wishes to account for the fact that the best observed model may have benefited from selection among many alternatives.

Another extension is the Model Confidence Set approach, which identifies a set of models that cannot be statistically distinguished as inferior under a chosen loss function. This would be conceptually attractive for the present research because the objective is not necessarily to identify a single universally dominant strategy. A set-based approach could instead characterize the strategies that remain statistically plausible competitors under a specified criterion.

However, formal model-selection procedures require careful choices about the loss function, dependence structure, and candidate-model set. Adding such methods without aligning them with the economic question could create the appearance of sophistication without improving identification. The existing framework therefore treats these procedures as extensions rather than substitutes for transparent baseline analysis.

Another possible extension is utility-based comparison. Rather than relying only on Sharpe ratios, an investor-specific utility function could combine expected return, variance, drawdown, or transaction costs. Such an approach would make the decision criterion explicit, but it would also require assumptions about investor preferences. The current multidimensional reporting avoids imposing those preferences.

A final extension is Bayesian portfolio construction, in which parameter uncertainty is represented directly in the posterior distribution of expected returns and covariances. This could provide a richer treatment of estimation risk but would

introduce prior distributions and additional modelling assumptions. The current study deliberately uses frequentist rolling estimates to maintain transparency and reproducibility.

These extensions illustrate the boundary of the present project. The core contribution is a clean empirical experiment, not an exhaustive survey of every portfolio-selection method. Keeping the primary design narrow helps ensure that each additional component can be audited and understood.

### 31. Practical Reproducibility Protocol for Independent Research

A reader attempting to reproduce the study should begin by obtaining the same source data or documenting any unavoidable differences in data availability. The retrieval environment should be recorded because financial-data vendors can revise historical series, change metadata, or alter the availability of older observations. The provenance file therefore records the retrieval context used by the computational run.

The next step is to install the dependencies from the project requirements file. Package versions should ideally be captured in the execution environment because numerical libraries can change optimization behavior or default data-processing conventions. The research pipeline should then be executed from the repository root so that generated output paths remain consistent.

After execution, the researcher should inspect the monthly return file and verify the date range and asset columns. The baseline performance file should contain four strategy rows. Robustness files should correspond to the predefined windows, costs, gamma values, covariance specification, and rebalance schedule. Missing or empty output files should be treated as failures requiring investigation.

The next audit step is to verify portfolio constraints. For every rebalance date and strategy, weights should sum to one within numerical tolerance and remain between zero and 0.30. If a strategy violates these restrictions, the optimization or post-processing logic should be examined before any results are interpreted.

The transaction-cost calculation should then be reconciled manually for a small number of observations. Gross return, turnover, cost rate, and net return should satisfy the stated accounting identity. This is a simple but powerful check because a sign or scaling error in basis points could materially alter conclusions.

Finally, the manuscript tables and figures should be populated only from generated output. The source code should remain the authoritative computational record, while the paper provides interpretation. This workflow reduces transcription errors and makes it possible to rerun the research after a code or data revision.

### 32. Research Contribution and Scope

The contribution of the project is best understood as an integrated empirical protocol. Many portfolio studies can be divided into separate questions about optimization, estimation, costs, or regimes. The present design places these elements in a common out-of-sample framework so that an observed performance difference can be examined from several angles.

The first contribution is methodological discipline around information timing. The rolling sequence explicitly prevents evaluation-period returns from entering the preceding portfolio decision. This is fundamental to credible backtesting and provides a clear distinction between a historical optimization exercise and a simulated investment process.

The second contribution is the common comparison set. Equal Weight, Minimum Variance, Mean-Variance, and Equal Risk Contribution are evaluated under shared investment constraints. This allows the incremental complexity of each method to be examined without changing the permissible leverage, shorting, or concentration structure.

The third contribution is implementation-aware evaluation. Turnover and transaction costs are incorporated directly into net returns, while cost sensitivity and rebalancing frequency are treated as predefined robustness dimensions. This recognizes that portfolio construction produces trades, not just mathematical weights.

The fourth contribution is explicit uncertainty analysis. Moving-block bootstrap intervals and pairwise Sharpe differences shift the discussion away from isolated point estimates. The design also acknowledges multiple-testing risk and avoids treating a large robustness grid as a source of unlimited confirmatory evidence.

The scope is intentionally limited. The project does not claim to discover a new asset-pricing factor, develop a superior forecasting model, or establish a universal optimal portfolio. Its purpose is to provide a transparent empirical test of whether additional portfolio-optimization structure produces durable historical evidence after estimation and implementation frictions are incorporated.

### 33. Identification Logic and Sources of Bias

The empirical design can be interpreted as an attempt to isolate the effect of portfolio construction while controlling several common sources of backtest bias. The most important is look-ahead bias, addressed by the strict t-minus-one information rule. A second is survivorship bias. The selected universe consists of currently identifiable ETFs rather than a historical database of all securities that existed at every date, so the study does not claim to eliminate survivorship effects in a broad security-selection universe. This limitation is explicit and should be considered when interpreting the scope of the evidence.

A third issue is data snooping. The project reduces this risk through a fixed strategy set and predefined robustness grid. This does not eliminate researcher discretion entirely, but it limits the ability to change the core experiment after observing results. The repository's methodology document is therefore part of the identification strategy: it records the intended rules before the empirical narrative is finalized.

A fourth issue is implementation leakage. If transaction costs are calculated after returns have already been compounded, or if portfolio weights are normalized using future observations, the apparent net performance can be distorted. The pipeline calculates costs at the monthly decision level and deducts them from the same period's gross return. This keeps the accounting sequence aligned with the simulated investment process.

A fifth issue is changing sample composition. The common seven-asset panel is kept constant so that a strategy is not evaluated on a different set of securities in different periods. This creates a transparent comparison but limits the generalizability of the result. The research therefore prioritizes internal consistency of the experiment over a claim of universal representativeness.

These design choices do not make the study immune to every bias. They establish a set of explicit assumptions under which the empirical results can be interpreted. The final paper should distinguish between biases addressed by construction and limitations that remain because of the available data and the chosen universe.

### 34. Relationship Between Statistical and Economic Models

The portfolio rule and the performance model answer different questions. The portfolio rule determines weights using historical estimates, while the performance model describes the realized return generated by those weights. It is possible for a mathematically well-defined portfolio rule to have weak out-of-sample performance because the statistical estimates entering the rule contain substantial error.

This distinction is visible in Mean-Variance optimization. The objective assumes that the sample mean and covariance provide useful information about future returns and risk. The empirical evaluation does not assume that premise is correct; it tests it. Minimum Variance performs a narrower test because it excludes expected returns. Equal Risk Contribution tests a risk-budgeting principle. Equal Weight provides the low-parameter benchmark.

The research can therefore be interpreted as a comparison of information requirements. A more complex model has more opportunities to exploit genuine structure but also more opportunities to exploit noise. Out-of-sample evaluation is the mechanism used to determine whether the additional information survives beyond the sample in which it was estimated.

The same logic applies to regime analysis. The regime label is not a forecast and is not used to choose weights in the baseline pipeline. It is an explanatory variable for conditional performance. This separation prevents the descriptive regime analysis from becoming an unplanned market-timing strategy.

The distinction between statistical and economic models also clarifies the role of transaction costs. Costs do not change whether an optimization problem is mathematically solved. They change whether the resulting portfolio is economically attractive to implement. The study therefore evaluates the optimizer first as a statistical decision rule and then as a trading process with explicit friction.

### 35. Expected Empirical Patterns and Falsification Criteria

The study does not embed empirical results in the manuscript, but the hypotheses imply observable patterns that can be used as falsification criteria. H1 would receive limited support if optimized strategies showed no meaningful improvement in risk-adjusted performance relative to Equal Weight across the baseline and robustness specifications. Conversely, a persistent improvement would require evidence across more than one specification.

H2 would be supported by meaningful changes in performance or portfolio composition across estimation windows. A result that remains broadly stable from 36 to 120 months would be less dependent on one information horizon. A result that changes materially would demonstrate estimation-window sensitivity and would require that qualification in the conclusion.

H3 can be examined directly through the cost grid. If a gross advantage remains almost unchanged as costs increase, turnover is not a major determinant of the relative result under the specified assumptions. If the advantage contracts quickly, implementation is an important part of the explanation. The relevant evidence is the complete cost-response curve rather than one selected cost point.

H4 concerns covariance shrinkage. Evidence consistent with the hypothesis would include reduced weight instability, lower turnover, or altered realized risk under the shrinkage specification. If shrinkage has little effect, that is also informative: it would suggest that the particular covariance-estimation adjustment does not materially change the experiment within this universe.

H5 concerns conditional market states. A strategy need not perform identically across regimes for the hypothesis to be informative. The important question is whether the differences are systematic enough, and sufficiently supported by the available observations, to characterize the historical behavior as regime-dependent rather than as isolated monthly noise.

These criteria emphasize that the research is capable of producing mixed evidence. A strategy can support one hypothesis while contradicting another. Such a result is not a methodological failure; it can identify the specific channel through which portfolio optimization appears useful or fragile.

## 36. Discussion of Model Complexity and Research Design

Complexity is an economic cost of a portfolio model even when it does not appear directly in the return series. A strategy that requires expected-return estimation, covariance estimation, numerical optimization, frequent rebalancing, and monitoring has greater operational requirements than Equal Weight. The project does not attempt to assign a monetary value to this complexity, but it measures several observable consequences through turnover, concentration, and parameter sensitivity.

The complexity argument also motivates the baseline strategy set. Equal Weight is deliberately simple; Minimum Variance adds one estimated object; Mean-Variance adds a second; Equal Risk Contribution changes the objective while retaining covariance estimation. This progression creates a natural laboratory for examining whether incremental modelling structure is associated with incremental empirical value.

A model should also be judged by stability. If a small change in a parameter causes a large change in weights, implementation may require continuous monitoring and discretionary intervention. A stable model can be easier to maintain even if its headline performance is not the highest point estimate. The robustness matrix and weight diagnostics provide evidence about this dimension.

The research design therefore treats complexity as something that must earn its place. An optimizer does not receive credit merely for producing a more sophisticated allocation. Its additional structure must correspond to an observable improvement in the out-of-sample evidence profile, or at minimum provide a risk or implementation characteristic that is economically interpretable.

This perspective is consistent with the project's broader research philosophy: the objective is not to maximize the apparent sophistication of the backtest. It is to create a transparent experiment in which each modelling choice has a defined purpose and each empirical claim can be traced to an output.

## 37. Final Audit and Submission Protocol

Before the manuscript is treated as final, the research should undergo a separate audit from the writing process. First, the title, author, date, research question, and strategy definitions should be checked for consistency across the manuscript and repository. Second, the baseline parameters should be reconciled with the code. Third, every numerical empirical claim should be traced to a generated result file.

The audit should also verify that no result table mixes gross and net returns, that transaction costs use the stated basis-point conversion, and that the regime signal is lagged. Portfolio weights should be tested against the full-investment and maximum-weight constraints. The first evaluation month should be checked manually to confirm that the estimation window ends before the evaluation return.

The PDF and DOCX should be inspected for formatting defects: broken equations, orphan headings, excessive blank space, missing page numbers, truncated tables, and inconsistent reference formatting. The goal is a compact academic layout in which page count comes from substantive content rather than artificial spacing.

The repository should then be checked for consistency between the paper and the source code. If the manuscript describes a method that the code does not actually implement, the manuscript must be corrected or the code upgraded before

submission. The same principle applies to robustness claims: a robustness experiment should not be described as completed unless its output exists and has been audited.

The final submission package should contain the manuscript, reproducibility documentation, methodology, bibliography, and source code. This provides a stronger record of independent research than a standalone PDF because it allows an evaluator to inspect the design and, where feasible, reproduce the computational process.

### 38. Internal Consistency Checks

Internal consistency is a separate dimension of research quality from the sophistication of the statistical method. The portfolio accounting identity should hold for every evaluation month: net return equals gross return less the transaction-cost deduction. The turnover measure should equal the sum of absolute changes in the relevant portfolio weights at each rebalance. These identities can be checked independently of the performance results and provide simple safeguards against implementation mistakes.

Weight constraints should also be tested at the observation level. The sum of weights should equal one within numerical tolerance, while every weight should remain within the zero-to-30% interval. If the optimizer returns an infeasible solution, the run should stop or record the failure rather than silently clipping the weights. Silent clipping would change the portfolio rule after optimization and make the empirical procedure difficult to reproduce.

The output structure provides another consistency check. Each baseline strategy should have the same evaluation dates, except where a computational failure is explicitly documented. Robustness files should change only the intended parameter. For example, a cost-sensitivity run should preserve gross returns and alter the transaction-cost deduction; a covariance-shrinkage run should preserve the strategy definition while changing the covariance estimator.

Finally, figures should be reconciled with tables. The terminal value shown in a cumulative wealth plot should correspond to the compounded net return reported in the performance table. Any discrepancy indicates that the plotting code and summary code are not using the same return series. These checks are mundane but important because errors at this level can invalidate otherwise sophisticated analysis.

### 39. Overall Research Design Synthesis

Taken together, the research design forms a sequence of linked tests. The data layer defines the investable universe. The estimation layer determines what information is available to the portfolio rule. The optimization layer converts that information into weights. The implementation layer converts weights into trades and net returns. The evaluation layer measures realized performance. The inference layer quantifies uncertainty. The robustness layer examines whether the interpretation survives changes in reasonable assumptions.

This sequence matters because a weakness in an earlier layer can propagate into later conclusions. An incorrectly aligned return series can produce incorrect covariance estimates; incorrect covariance estimates can produce incorrect weights; incorrect weights can produce incorrect turnover; and incorrect turnover can produce incorrect net returns. The research therefore treats the computational pipeline as a chain in which each stage should be auditable.

The design also creates a clear distinction between what the research can establish and what it cannot. It can establish the historical behavior of the specified portfolio rules under the stated assumptions. It can identify whether relative performance is sensitive to costs, windows, covariance estimation, rebalancing, or market regimes. It cannot establish that a historical advantage will persist indefinitely or that an investor with different constraints will obtain the same outcome.

For an MSc-level research project, this distinction is important. Academic quality is not created by adding complexity for its own sake. It is created by posing a precise question, defining a credible identification strategy, using appropriate statistical methods, documenting limitations, and ensuring that the conclusion follows from reproducible evidence. The present design is structured around those principles.

### 40. Closing Statement on Empirical Interpretation

Once the computational pipeline has been executed and audited, the empirical results should be inserted into the designated tables and figures without changing the underlying methodological rules. The narrative should then be rewritten where necessary to describe the observed patterns precisely. If the evidence contradicts an initial hypothesis, the hypothesis should be reported as unsupported or only partially supported rather than being reinterpreted after the fact.

The final discussion should give particular attention to whether apparent optimization benefits survive the transition from gross to net performance. It should also examine whether the same pattern appears under alternative estimation windows and covariance estimators. These tests are central because they address the two principal practical objections to optimization: that its inputs are noisy and that its implementation can be costly.

The strongest empirical contribution will therefore come from consistency across evidence types. A return difference supported by stable weights, moderate turnover, persistence across windows, and uncertainty intervals that exclude economically negligible differences would constitute a different evidence profile from a return difference accompanied by concentration, high turnover, and strong specification dependence. The paper's multidimensional framework is designed to make these distinctions explicit.

The final conclusion should remain a statement about the evidence generated by the experiment. It should not become a forecast, an investment recommendation, or a claim of universal optimality. The value of the project lies in showing, with a transparent and reproducible process, what happens when portfolio optimization is evaluated under the statistical and economic conditions that real investment decisions must confront.

## 41. Researcher Degrees of Freedom and Transparency

Even a carefully specified backtest contains discretionary choices. The choice of assets, sample period, estimation window, transaction-cost grid, performance metrics, regime definition, and statistical procedures can all affect the empirical record. Transparency does not require pretending that these choices are uniquely correct. It requires identifying them, explaining their purpose, and avoiding the selective presentation of whichever choice produces the most favorable result.

The present project addresses this issue by separating baseline decisions from robustness decisions. The baseline is fixed as a 60-month window, monthly rebalancing, 15 basis points of transaction costs, gamma equal to 3, sample covariance, long-only full investment, and a 30% maximum weight. Alternative values are then examined through a predefined grid. This structure allows a reader to distinguish the principal experiment from sensitivity analysis.

The same transparency principle applies to negative or ambiguous findings. If two strategies have statistically indistinguishable Sharpe ratios, the paper should report that fact rather than infer a meaningful difference from small decimal variations. If a strategy's performance changes materially across specifications, that instability is itself a result. Treating uncertainty as evidence rather than as an inconvenience improves the credibility of the research.

In this sense, reproducibility is not simply the ability to rerun code. It is the ability to reconstruct the logic connecting a research question to a data set, a decision rule, an implementation assumption, a statistical measure, and a conclusion. The repository and manuscript are designed to preserve that chain.

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